DRUGS · ANTIBIOTICS: PENICILLINS

Flucloxacillin

Flucloxacillin is a penicillinase-resistant penicillin: its side chain shields the beta-lactam ring from staphylococcal penicillinase. It is the drug for infections caused by Staphylococcus aureus: furuncles of the ear canal and nasal vestibule, cellulitis of the face and pinna, infected wounds and haematomas, and acute suppurative parotitis. It does not cover MRSA, Pseudomonas or Haemophilus.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

FLUCLOXACILLIN

	SUMMARY
Class	Penicillinase-resistant (antistaphylococcal) penicillin
How it works	Cell wall synthesis inhibitor, stable against staphylococcal penicillinase
Spectrum	Methicillin-sensitive S. aureus, streptococci; not MRSA, not Gram-negatives, not Pseudomonas
ENT uses	Furunculosis of the ear canal and nasal vestibule; cellulitis of the face, pinna and neck; infected post-operative wounds; acute suppurative parotitis (with an anaerobic or broader agent if unwell); infected auricular haematoma; impetigo; with benzylpenicillin for severe cellulitis
Adult dose	Oral 500 mg to 1 g four times daily; intravenous 1 to 2 g four times daily (up to 2 g every 4 hours in endocarditis or osteomyelitis)
Child dose	Oral 1 month to 1 year 62.5 to 125 mg, 2 to 9 years 125 to 250 mg, 10 to 17 years 250 to 500 mg, four times daily; intravenous 25 to 50 mg/kg four times daily
Duration	5 to 7 days for cellulitis, by response
Avoid	Penicillin allergy; previous flucloxacillin-associated jaundice

	SUMMARY
Pregnancy	Safe

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

An isoxazolyl penicillin whose bulky side chain prevents penicillinase from reaching the beta-lactam bond. **MRSA** resists it through a different penicillin-binding protein (PBP2a, encoded by mecA), so methicillin resistance means flucloxacillin resistance.

2.2 Absorption and handling

Oral absorption is incomplete and reduced by food: take it **an hour before or two hours after meals**. It is highly protein-bound, excreted renally and partly by the liver.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Staphylococcal skin and soft tissue infection

- **Furuncle of the ear canal**: a painful boil in the outer, hair-bearing canal; incise if pointing, give oral flucloxacillin for spreading infection, and a wick with antibiotic-steroid drops if the canal is shut.
- **Nasal vestibulitis and furunculosis**: topical antiseptic or mupirocin for mild disease, oral flucloxacillin for a furuncle, and urgent assessment if there is spreading cellulitis (the danger triangle drains to the cavernous sinus).
- **Cellulitis of the face, pinna and neck**: flucloxacillin, with benzylpenicillin or

clindamycin in severe disease.

3.2 Perichondritis of the pinna

Perichondritis after piercing of the upper ear cartilage is often caused by **Pseudomonas**, which flucloxacillin does not cover; ciprofloxacin is the oral choice. Use flucloxacillin for simple cellulitis of the lobule and staphylococcal wound infection.

3.3 Acute suppurative parotitis

Usually S. aureus in a dehydrated, elderly or post-operative patient: rehydrate, massage, sialogogues and intravenous flucloxacillin (with metronidazole or as co-amoxiclav), and look for an abscess on ultrasound if it does not settle.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Cholestatic jaundice and hepatitis**, which may appear up to two months after stopping; commoner in the elderly and with courses longer than 14 days.
- Gastrointestinal upset; phlebitis with intravenous use.
- The hypersensitivity of all penicillins.

4.2 Contraindications and cautions

- Penicillin allergy; previous flucloxacillin jaundice.
- Hepatic impairment.

4.3 Interactions

Paracetamol with flucloxacillin, in malnourished or septic patients, has been linked to high anion gap **metabolic acidosis** (pyroglutamic acidosis). Methotrexate excretion is reduced.

CHAPTER FIVE

5 For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Why flucloxacillin resists penicillinase, and why MRSA resists flucloxacillin.
- The organisms of **otitis externa (Pseudomonas, S. aureus)** and of **perichondritis (Pseudomonas)**, and which drug covers which.
- Take on an **empty stomach**; delayed cholestatic jaundice.

5.2 Common pitfalls

COMMON PITFALLS

- Treating a cartilage perichondritis with flucloxacillin alone.
- Oral flucloxacillin taken with food.
- Forgetting to drain a pointing furuncle or an abscess.

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